Extra Dessert #2: Noise and Interference

DNR, noise blankers, ANF, noise source recognition, common-mode current, and station grounding.

Group E4E · 14 pool questions · Self-study reference · Practice exams at eham.net/exams

Key Facts Reference

Digital Noise Tools And Their Side Effects

Tool	Best at removing	Watch out for
Digital noise reduction (DNR)	Broadband white noise, ignition noise, power-line noise – **all of the above**. (E4E02)	—
Noise blanker	**Impulse noise** (sharp pulses). (E4E03)	Strong signals may be distorted and appear to cause spurious emissions. (E4E09)
Automatic notch filter (ANF)	Carriers / steady tones	Can **remove the CW signal as well as the interfering carrier**. (E4E01) Turn off ANF when working CW.

Noise Source Recognition

Noise signature	Source	Q
Series of carriers at regular intervals across a wide frequency range	**Switch-mode power supplies**	E4E12
Unstable modulated or unmodulated signals at specific frequencies	**Computer network equipment**	E4E06
Intermittent loud roaring/buzzing on AC line	Arcing contacts in a thermostatically controlled device, defective doorbell or doorbell transformer, malfunctioning illuminated advertising display – **all of the above**	E4E10
Local AM broadcast stations mixing into MF/HF bands	**Nearby corroded metal connections mixing and reradiating the broadcast signals** (the “rusty bolt effect” – mixing is what produces the spurious sum/difference signals)	E4E11

Conducted Noise Fixes

Source	Fix	Q
Automobile battery charging system	**Ferrite chokes on the charging system leads**	E4E04
Line-driven AC motor	**Brute-force AC-line filter in series with the motor’s power leads**	E4E05

Common-Mode Current and Shield Behavior

Question	Answer	Q
Current that flows equally on all conductors of an unshielded multiconductor cable	**Common-mode current**	E4E08
Why shielded cables radiate or receive interference	**Common-mode currents on the shield and conductors**	E4E07

Grounding and Surge Protection

Question	Answer	Q
Where to install an AC surge protector	**On the single point ground panel** – not the AC service panel or a random outlet	E4E13
Purpose of a single point ground panel	**Ensures all lightning protectors activate at the same time**	E4E14

Dessert 2 Question Bank (E4E)

E4E01. What problem can occur when using an automatic notch filter (ANF) to remove interfering carriers while receiving CW signals?
A. Removal of the CW signal as well as the interfering carrier.
B. Any nearby signal passing through the DSP system will overwhelm the desired signal.
C. Excessive ringing.
D. All these choices are correct.

E4E02. Which of the following types of noise can often be reduced by a digital noise reduction?
A. Broadband white noise.
B. Ignition noise.
C. Power line noise.
D. All these choices are correct.

E4E03. Which of the following types of noise are removed by a noise blanker?
A. Broadband white noise.
B. Impulse noise.
C. Hum and buzz.
D. All these choices are correct.

E4E04. How can conducted noise from an automobile battery charging system be suppressed?
A. By installing filter capacitors in series with the alternator leads.
B. By installing a noise suppression resistor and a blocking capacitor at the battery.
C. By installing a high-pass filter in series with the radio's power lead and a low-pass filter in parallel with the antenna feed line.
D. By installing ferrite chokes on the charging system leads.

E4E05. What is used to suppress radio frequency interference from a line-driven AC motor?
A. A high-pass filter in series with the motor's power leads.
B. A brute-force AC-line filter in series with the motor's power leads.
C. A bypass capacitor in series with the motor's field winding.
D. A bypass choke in parallel with the motor's field winding.

E4E06. What type of electrical interference can be caused by computer network equipment?
A. A loud AC hum in the audio output of your station's receiver.
B. A clicking noise at intervals of a few seconds.
C. The appearance of unstable modulated or unmodulated signals at specific frequencies.
D. A whining-type noise that continually pulses off and on.

E4E07. Which of the following can cause shielded cables to radiate or receive interference?
A. Low inductance ground connections at both ends of the shield.
B. Common-mode currents on the shield and conductors.
C. Use of braided shielding material.
D. Tying all ground connections to a common point resulting in differential-mode currents in the shield.

E4E08. What current flows equally on all conductors of an unshielded multiconductor cable?
A. Differential-mode current.
B. Common-mode current.
C. Reactive current only.
D. Magnetically-coupled current only.

E4E09. What undesirable effect can occur when using a noise blanker?
A. Received audio in the speech range might have an echo effect.
B. The audio frequency bandwidth of the received signal might be compressed.
C. Strong signals may be distorted and appear to cause spurious emissions.
D. FM signals can no longer be demodulated.

E4E10. Which of the following can create intermittent loud roaring or buzzing AC line interference?
A. Arcing contacts in a thermostatically controlled device.
B. A defective doorbell or doorbell transformer inside a nearby residence.
C. A malfunctioning illuminated advertising display.
D. All these choices are correct.

E4E11. What could be the cause of local AM broadcast band signals combining to generate spurious signals on the MF or HF bands?
A. One or more of the broadcast stations is transmitting an over-modulated signal.
B. Nearby corroded metal connections are mixing and reradiating the broadcast signals.
C. You are receiving skywave signals from a distant station.
D. Your station receiver IF amplifier stage is overloaded.

E4E12. What causes interference received as a series of carriers at regular intervals across a wide frequency range?
A. Switch-mode power supplies.
B. Radar transmitters.
C. Wireless security camera transmitters.
D. Electric fences.

E4E13. Where should a station AC surge protector be installed?
A. At the AC service panel.
B. At an AC outlet.
C. On the single point ground panel.
D. On a ground rod outside the station.

E4E14. What is the purpose of a single point ground panel?
A. Remove AC power in case of a short-circuit.
B. Prevent common-mode transients in multi-wire systems.
C. Eliminate air gaps between protected and non-protected circuits.
D. Ensure all lightning protectors activate at the same time.

Answer Key

E4E01: A	E4E02: D	E4E03: B	E4E04: D
E4E05: B	E4E06: C	E4E07: B	E4E08: B
E4E09: C	E4E10: D	E4E11: B	E4E12: A
E4E13: C	E4E14: D		